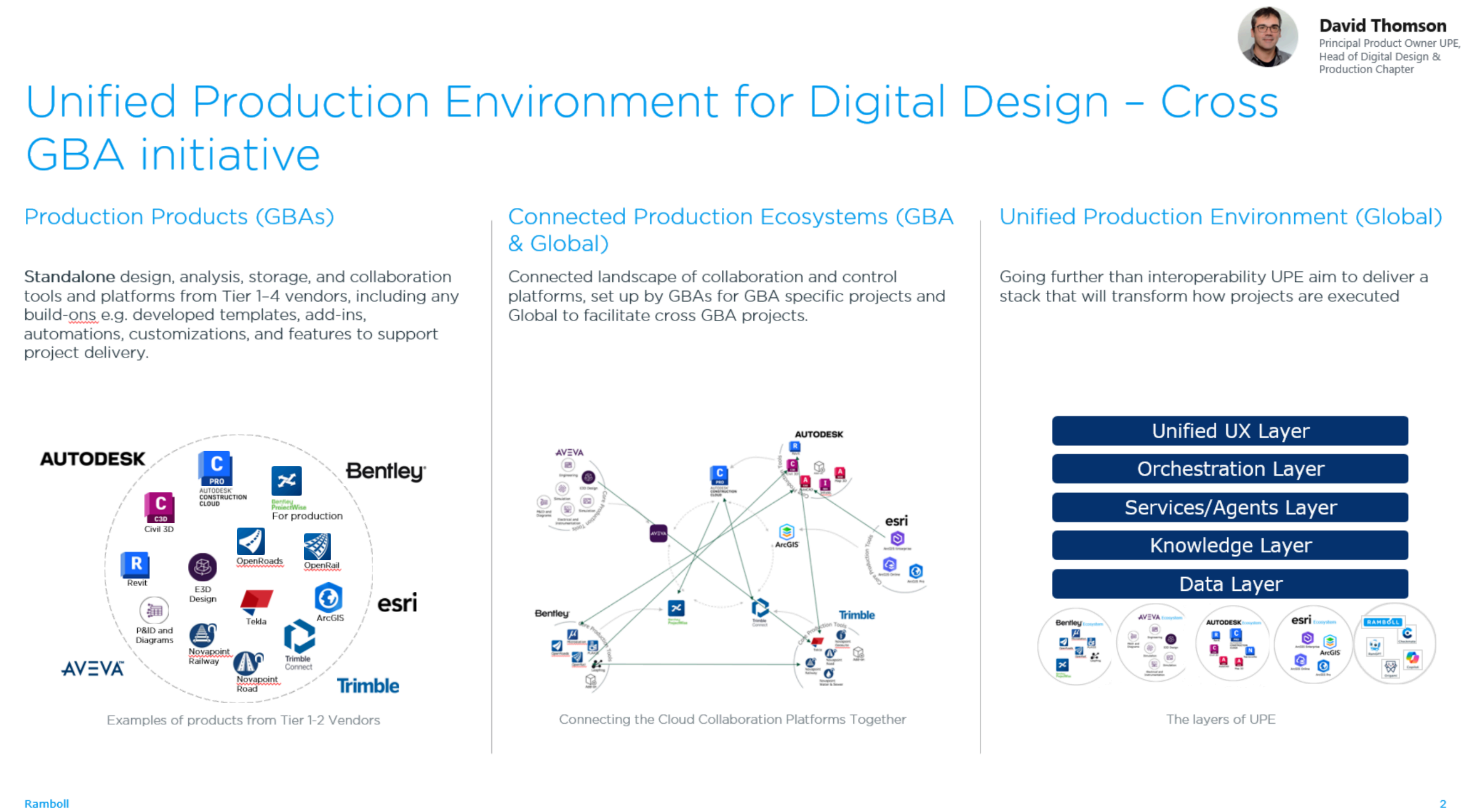
Summary of UPE / DDP Kickoff + Strategic Observations

Fresh thoughts (David Ts 3rd March)

- Historically projects were executed with desktop products, which were used primarily in a standalone way , i.e no underlying platform like ACC (which is probably still the case for the very small projects)
  - There are too many of these stand alone products (IDTs are harmonizing them)
  - Customizations /Automations were local or at least locally administrated
  - Files were or are still stored locally on file systems or in a non standardized way (Suspect Sharepoint)
- Larger projects and geographies dealing with higher value projects had large incentives to move to collaboration and control platforms (CDE+) like ACC. These platforms solve many of the basic issues of coordinating work across disciplines and teams, common file storage and status, centrally managed workflows, common place for clash , interface and design reviews etc.
- When a GBA needs to work outside of a single platform ecosystem the old issues of file management, file conversions and workflows reappear.
- Or when a GBA is working on a cross GBA project where again there is no single CDE+ platform file flow and workflows need to be manually managed. **This is the immediate UPE Gap - file flow and orchestration**
- Moreover the CDE+ platforms from Autodesk, etc are relatively simple growing up from simple file management to basic collaboration platforms, where ACC is rapidly maturing with visualization etc. (Project Wise might be more advanced , please

confirm Eirik) I suspect these CDE+ platform do not have data centric project execution capabilities such as data standards, data decomposition/scraping/normalization capabilities that would be needed to work in fully data centric ways which would be needed to exploit AI fully and to amongst others move into new business areas such as operational services. **This is the UPE vision**

I came up with this new graphic for UPE for the RBU Tech connect day. Connected Production Ecosystems is phase 1 where we are today with UPE, the layers diagram is the vision .....



1. Observations from the UPE & DDP Kickoffs

1.1. Alignment Challenge on the Name "UPE"

- The term **Unified Production Environment** has the potential to trigger confusion at the global level because:
    - Buildings IDT already use terms like **Unified Digital Delivery (UDD)** and **Unified Framework**.

- Henning Larsen have their own "UPE" due to their single-discipline setup.
- This led to misalignment on what "UPE" means globally vs. within a GBA.
- See recommendations below

1.2. Diverse Requirements Across GBAs

- **Henning Larsen**
  - Already built a lightweight UPE equivalent as they are (mostly) single discipline and heavily Autodesk based, their UPE is essentially ACC.
  - Thier global UPE requirements are interoperability related: i.e. **i.e. how they do data exchange with other GBAs in a cross disciplinary project.**
- **Buildings**
  - Have their own **Unified Digital Delivery** strategy focused on software harmonization onto the ACC platform, they recognize the need for better data and information management to deliver digitially ready projects and add value to their service lines.
  - Generally RBU need **the interoperability layer** with other platforms and the **knowledge repository** at the Global UPE level.
- **Transport, Water, E&H**
  - Smaller or fragmented IDT teams.
  - Much **greater reliance on Global** to deliver an end to end UPE.
  - Transport especially needs a **robust multi-discipline ecosystem with data management**.
  - E&H are more document centric - can the new DMS fulfill some of those requirements?

These differences make it clear that **UPE means something different in each GBA**.

1.3 AI Discussion – Two-Strategy Approach

- General consensus:
  - **Bottom-up innovation** on projects should be allowed and encouraged, we discussed a MVP from Global is a database of existing Citizen developer AI tools and tech stack requirements/templates. In work with the Product team
  - **Top-down enablement** needed for vendor tools (Autodesk etc.) each vendor is rolling out chatbots, and MCP infrastructure , RT needs to ensure a structured roll out with guide rails.
- Agreement that guardrails are needed quickly, but no formal decisions yet.

1.4 Data & Information Management Escalated

- UPE includes a data workstream, but maturity is low in Ramboll Tech and some GBAs.
- Buildings IDT are hiring an **Information Management Specialist** from Mott.
- Business Excellence hiring a **Solution Architect** with potential DDP background.
- Agreement in principle to establish a **Data & Information Management Chapter**.
- David T is **currently the provisional lead**.

1.5 Production Standards and Processes

Processes and production standards are a very important part of delivering UPE, they are the map and will steer some of the capabilities going forward either in controlled work flows or via agents. As far as I understand many of our production processes are documented in Camino but many remain documented in inconsistent ways or remain as tribal knowledge. It important we map out the most important UPE relevant processes as a minimum. I don't have a good overview of where we are with this. However its clear :

- we need to make it easier to capture this kind of knowledge
- we need to document this in a machine /AI friendly way that is consistent across the business
- we need to think about how these processes are surfaced / exploited by the users.

2. Recommendations (proposals, not meeting outcomes)

2.1 New Naming: Shift from UPE → Unified Project Execution (low priority)

Your proposal:

- Global level should adopt: **Unified Project Execution (UPE)**

(clearer scope: covers interfaces with business systems, project controls, administrative tools).

(Expect push back on this as RT leadership want more focus on the production part of UPE to keep it focused.)

- GBAs can define their own **Unified Digital Delivery** strategies but ideally R Tech keep the overview to avoid silos and overlaps.
  - This accommodates major differences between GBAs.
  - Reduces naming conflicts (e.g., Buildings' existing UDD strategy).

2.2 Standardizing GBA Requirement Documentation (low priority)

- Each GBA currently maps its:
  - software ecosystem,
  - workflows/processes,
  - tool-to-process mapping—

but **in inconsistent ways**.

- Recommendation:
    - Each GBA should document **software ecosystems per major service line**.
    - Aim for a consistent format across GBAs for comparability.
    - Include **AI tool usage** in this mapping.

2.3 Enabling Connected Production Ecosystems (High priority)

See top of this page **This is the immediate UPE Gap - file flow and orchestration**

We need to map the needed file flows and the solutions or gaps ASAP, this is phase 1 of UPE.

2.4 AI Governance – Bottom-Up + Top-Down (High Priority)

Your recommended model:

- **Bottom-Up**
    - Allow project-level AI innovation (vibe coding, ad hoc automation, rule checking etc.).
    - Avoid stifling productivity; embrace the “citizen developer” era.
    - Introduce minimum viable guardrails:
        - **Database** of solutions built across the business
        - **Recommended tech stack templates**
        - **Rules for project ownership**
        - Optional **escalation into Tech Governance → DPE** if scalable globally
- **Top-Down**
    - Ensure broad enablement of vendor AI tools (e.g., Autodesk).
    - Deliver training and structured adoption support.

2.5 Unified Data Strategy (High Priority)

- Recommendation:
  - Consider **Engineering/Production Data** and **Enterprise Data** together, not separately.
  - Needed due to emerging AI use cases across all types of data.
  - Global role required: **Owner of Data Strategy for DDP/Engineering**.

2.6 Unified Process Strategy (Medium Priority)

- Recommendation:
  - Consider **Engineering/Production Processes** and **Enterprise Processes** together, not separately.
  - Needed due to emerging AI use cases across all types of processes
  - Business excellence and UPE Process Owner (Diana) meeting in next few weeks to align

2.6 Capability-Based lens on UPE and DD&P (Not Tool-Based)

- We collectively need to move from tool ownership → capability leadership:
  - Geospatial chapter / capability lead
  - BIM/ 3D data centric design Chapter /capability lead
  - Processes and Knowledge Chapter / lead
  - Data and Information Management Chapter / lead
  - AI Chapter / Lead
- Each capability needs a strategy /vision /Rd Map which in turn gets turned in to deliverables via product managers.

3. Suggested Actions

3.1 Governance Work stream

3.1.0 Governance basics (David)

- **Set up Steer co**
- **Determine prioritization metrics for UPE backlog**
- **Set up backlog and single source of truth**

3.1.1 Naming & Scope (David)

- Stick with Unified Production Environment in the short term.
- Align with Buildings IDT on language in general, so their UDD framework remains local/GBA-specific.
- Socialize this framing with DEJ, and other enabling functions.
- Develop proposal for **Unified Project Execution** as the global concept.

3.1.2 Investigate reorganizing into capability chapters for DD&P Global Team (David to lead and align with IDT Tech leadership)

- Refine structure where each portfolio manager becomes a **chapter/capability lead**.
- Compliment the core team with chapter/capability leads for emerging capabilities i.e. Computational/generative design, Digital Twins etc.
- Document chapter responsibilities

- Plan transition away from tool-ownership mindsets.

3.1.3. Identify pilot projects to validate UPE concepts (Heads of)

3.2 Connected Production Ecosystems workstream

- Audit recent cross GBA projects to map current manual file routing steps and identify the most common patterns. (Michael and Diana)
- Establish export and import standards, is this IFC? (Michael, and Heads of)

3.3 Data & Information Management workstream

- Formalize the **Data & Information Management Chapter**. (David)
- Identify distributed experts across GBAs to participate. (Heads of DD&P)
- Define a mandatory data repository standard for all projects under a certain size - goal not to lose value from the data on those small projects and make sure its AI ready. This could be as simple as Sharepoint with a naming convention or folder structure. (IDT)
- Define the data model for UPE

3.4 Process and Knowledge workstream

- Formalize the way Production Processes and Business processes are captured , stored and utilized. **(Diana)**
- Create a **template for software ecosystem and process mapping**.
- Ask each GBA to provide:
  - their service-line workflows
  - mapped tools
  - AI tools currently in use

- future AI directions .

3.5 AI in DD&P work stream

- Produce initial citizen developer **Governance Starter Pack**: **(David and Robins team)**
  - How to register solutions
  - What tech stack to use
  - When to escalate to global
  - Project-level lifecycle responsibilities etc
- Plan cross-GBA forum to align on bottom-up practices.
- Align RamGPT team on potential use in UPE (Services /agents layer above)

3.6 Platform and Tech Enablement work stream

- Identify the gaps between the vendor supplied CDE+ platforms and the UPE vision (David and Heads of)
- Define a technical solution architecture for UPE (David and DPE)
- Determine buy build or partner for each of the major capabilities needed

3.7 Digital Employee Journey

- Initial planning and visibility for DEJ team. (David)